

# SAILAKSHMI B T

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## PROFESSIONAL SUMMARY

Robotics and Automation undergraduate (CGPA 8.99) at Madras Institute of Technology with hands-on experience in embedded systems, robotics, and AI-driven applications. Skilled in Arduino-based system design, sensor interfacing, and rapid prototyping. Interests span Computer Vision, industrial automation, embedded intelligence, and intelligent systems.

## EDUCATION

### Bachelor of Engineering – Robotics and Automation

Madras Institute of Technology | 2024 – 2028 | CGPA: 8.99 / 10.0

- Higher Secondary Certificate (2024) – 94.16%
- Secondary School Certificate (2022) – 95.2%
- JEE (Main) 2024 – Paper 1 (B.E./B.Tech.): 81.82 Percentile (NTA Score)

## INTERNSHIP & TRAINING

### Computer Vision & AI Summer Intern – PixelBotix

June 2026

- Developed practical skills in Computer Vision, image processing, object detection, and AI-based visual recognition through intensive hands-on training and worked on some basic projects.

### Industrial Robotics & Automation Training – Birla Carbon & KUKA

February 2026

- Gained exposure to industrial robotic arm programming, Industry 4.0 concepts, automation workflows, and industrial safety practices.

## PROJECTS & RESEARCH

### Capstone Project: Student Life Concierge AI Agent (Kaggle & Google 5-Day AI Agents Intensive)

Concierge Agents Track - Submission Grade | Google Agent Development Kit (ADK), Gemini API, Python, FastAPI, Pytest, Uvicorn

- Developed a multi-agent FastAPI system to automate academic scheduling via email parsing.
- Secured the agent with an anti-jailbreak layer.
- Established a comprehensive Pytest suite with mock services for local evaluation.

### Agentic Masters Application Journey Tracker & Resume Analyzer

React, Vite, LocalStorage API, PDF.js, CSS3 | In Progress

- Designing a 100% client-side, privacy-first web app to manage the Masters application pipeline.
- Building an offline Resume Intelligence Engine via pdfjs-dist for keyword extraction.
- Developing an ATS Scoring & Gap-Analysis Algorithm for compatibility scores and milestones.
- Implementing LocalStorage partitioning and a research planner for academic venues.

### Orion Local AI Assistant

Llama 3.2, Python, WebSockets, Web Speech API | In Progress

- Engineering a fully offline, privacy-first AI assistant integrating Meta's Llama 3.2 with Ollama.
- Building an asynchronous FastAPI backend with WebSocket communication for real-time agent state transitions.
- Implementing a function-calling framework enabling the AI to autonomously run live CPU/RAM diagnostics.

### Mechanical Rotary Device [Design Patent Pending]

- Designed a dual-mode rotary mechanism integrating a Geneva wheel and dog clutch system using SolidWorks; filed for a design patent.

### Grip Stability & Stress Analyzer

- Developed a load cell-based system using HX711 and Arduino Uno to evaluate grip stability and tremor patterns for stress-level estimation.

### Adaptive Self-Mutating Authentication System

- Designed an 8051-based digital lock using hardware timers to generate non-deterministic passwords after each login, mitigating replay attacks and shoulder-surfing.

Single-Axis Gimbal System

- Engineered a stabilization gimbal using Arduino Nano, MPU-6050, and SG90 servo motor, achieving real-time angular stability through inertial sensing and closed-loop feedback.

HACKATHONS & COMPETITIONS

Smart India Hackathon 2025 - Finalist

- Contributed to AI/ML-based identification of cryptographic primitives and protocols in multi-architecture firmware binaries.

H2TS Hackathon 2025 - 2nd Place

- Collaborated to develop an AI-driven SOS platform for emergency situations.

VISA 24 Hours Hackathon 2026 - Participant

- Competed in a 24-hour innovation sprint.

TECHNICAL SKILLS

**Control systems:** PID Control, Motion Control, Stabilization Mechanisms, Closed-Loop Feedback

**Programming & Analysis:** Python (Intermediate), Embedded C (Fundamentals), MATLAB/Simulink (Intermediate)

**Embedded Systems:** 8051 Architecture (AT89S52), Arduino Uno R3, Arduino Nano, Sensor Interfacing, Timer-Based Logic

**AI & Agentic Systems:** Google Agent Development Kit (ADK), Gemini API, Ollama, Llama 3.2, FastAPI, Pytest, Uvicorn, Prompt Engineering

**Web & Real-Time Systems:** WebSockets, Asynchronous Programming, Speech Recognition & Synthesis (Web Speech API)

**Design Tools:** SolidWorks, KiCAD, Canva

**Cybersecurity & RE:** Binwalk, Ghidra, Ubuntu, Firmware Binary Analysis (working knowledge of tools and workflow)

**Soft Skills:** Communication, Anchoring, Content Creation, Critical Thinking, Problem Solving

CERTIFICATIONS

- AI Capabilities and Limitations – Anthropic
- Foundations of Prompt Engineering – AWS Training & Certifications
- Python for Absolute Beginners – Udemy
- Basics of HTML – Udemy
- Automation in Manufacturing – NPTEL Elite Certification (67%)

LEADERSHIP & EXTRACURRICULARS

|                                                                                                        |                |
|--------------------------------------------------------------------------------------------------------|----------------|
| <b>Placement Representative</b>                                                                        | 2026 – Present |
| • Serving as liaison between students and placement coordinators for placement-related communications. |                |
| <b>YRC Volunteer</b>                                                                                   | 2024 – Present |
| • Helped organize Youth Fest events for visually challenged individuals.                               |                |
| <b>Member, ARME</b>                                                                                    | 2024 – Present |
| • Association of Robotics and Mechatronics Engineers.                                                  |                |
| <b>Joint Treasurer, MIT Quill</b>                                                                      | 2025 – 26      |
| • Managed treasury operations for the college literary club.                                           |                |
| <b>Office Bearer, SAHA</b>                                                                             | 2025 – 26      |
| • Science and Humanities Association — organized and coordinated academic events.                      |                |

LANGUAGES

English (Proficient) | Tamil (Proficient) | Telugu (Conversational) | Japanese (Elementary) | Hindi (Elementary)